SAPM Assignment - Summer Term

by

Arnav Agarwal (2022B1A41724P)
Atharva Toshniwal (2022B3A80224P)

Prepared for the fulfillment of the course

Securities Analysis and Portfolio Management (ECON F412)



Birla Institute of Technology and Science Pilani, Pilani Campus

5 July, 2025

DECLARATION

We, the members of this group, hereby submit our original project report on the assigned topic to partially fulfill the requirements of the ECON F412 course (Securities Analysis and Portfolio Management). We confirm that all of the members have contributed equally to the project. We share equal responsibility for the content presented, affirming it is plagiarism-free. All necessary references have been appropriately cited within the text.

GROUP MEMBERS

Arnav Agarwal	2022B1A41724P
Atharva Toshniwal	2022B3A80224P

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to our Instructor-in-Charge, **Professor C. Hussain Yaganti**, from the Department of Economics and Finance, BITS Pilani Hyderabad Campus for his invaluable guidance and support throughout the course **Securities Analysis and Portfolio Management.**

Professor Yaganti's profound expertise in the areas of securities and portfolio management significantly deepened our understanding of the subject. His engaging lectures, insightful discussions, and practical approach to complex financial concepts greatly enhanced the quality of our learning experience and the depth of our analysis.

We are especially thankful for his patience in addressing our queries, his thoughtful suggestions that helped us explore alternative perspectives, and his consistent encouragement, which motivated us to push our boundaries. His ability to simplify intricate topics and relate them to real-world applications has not only clarified our doubts but also inspired us to think critically and analytically.

This journey of preparing our report under his mentorship has been both enriching and transformative. We are truly grateful for the knowledge, skills, and academic rigor we have gained under his guidance. The experience has been instrumental in shaping our understanding of securities analysis and portfolio management, and it will undoubtedly serve as a strong foundation in our future academic and professional endeavors.

Index

---

1. Abstract
2. Approach Summary
    a. Our Approach
    b. Industry 1: Tele - Communications
    c. Industry 2: Defence
    d. Industry 3: IT
    e. Industry 4: FMCG
    f. Industry 5: Banks
    g. XAU/INR – Gold Price per Gram (INR):
3. Assignment Part A: Individual Stocks Portfolio
4. Assignment Part B: Stocks Back-Testing
5. Assignment Part C: Indices Portfolio
6. Part D: Indices Back-Testing
7. Part E: Final Conclusions
    a. Observations
    b. Conclusions
8. Tools and Sources Used

Abstract

This study evaluates the construction, optimization, and performance of investment portfolios comprising individual domestic stocks and broad market indices within the Indian financial landscape. Conducted as part of the course *Securities Analysis and Portfolio Management*, the research applies empirical methods to analyze return, risk, and efficiency of portfolios over a one-year investment horizon, followed by short-term back-testing for robustness.

The analysis is structured around four key components: (A) selection and justification of four domestic stocks based on qualitative and quantitative criteria; (B) portfolio construction using equal weights, with subsequent optimization through Excel Solver to maximize the Sharpe ratio; (C) one-month back-testing of the optimized stock portfolio to assess short-term alignment with historical performance; and (D) replication of the same process using four domestic indices to explore diversification at the macro-market level.

Findings reveal that while equal-weighted portfolios offer simplicity and initial balance, optimization significantly improves risk-adjusted returns. The back-testing exercise highlights key differences between theoretical efficiency and real-world performance, emphasizing the impact of volatility and short-term market movements. The report concludes with a synthesis of investment outcomes and offers strategic recommendations for portfolio structuring, including asset-class diversification, active monitoring, and periodic rebalancing for sustainable, data-driven investment management.

These insights are applicable to academic, retail, and emerging investors seeking to bridge the gap between textbook theory and market practicality in a structured, quantifiable manner.

Approach Summary

Our Approach:

Step 1: Industry Identification

We began by identifying key industries based on their overall economic importance, market stability, and historical return potential. Priority was given to sectors that demonstrated strong growth prospects and consistent financial performance.

Step 2: Company Shortlisting

Within each selected industry, five leading companies were shortlisted. The selection was based on criteria such as market capitalization, brand strength, historical returns, and the company's prominence and relevance within its sector.

Step 3: Evaluation Framework Development:

To evaluate each shortlisted company, we developed a structured framework comprising five core metric categories:

Valuation metrics

1. **P/E Ratio (Price-to-Earnings Ratio):**

This ratio indicates how much investors are willing to pay for ₹1 of a company's earnings. A high P/E suggests high future growth expectations, while a low P/E may indicate undervaluation or poor growth prospects. It's most useful when comparing companies within the same industry.

Formula: Market Price per Share / Earnings per Share (EPS)

Interpretation: A **lower** P/E is generally preferred, unless a high value is supported by strong, consistent earnings growth.

2. P/B Ratio (Price-to-Book Ratio):

The P/B ratio compares a company's market value to its book value (net assets). A ratio above 1 implies the stock is trading at a premium to its book value, potentially due to investor confidence, while a ratio below 1 could signal undervaluation or financial concerns.

Formula: Market Price per Share / Book Value per Share

Interpretation: A **lower** P/B ratio is typically preferred, especially when supported by solid fundamentals.

3. EV/EBITDA (Enterprise Value to Earnings Before Interest, Taxes, Depreciation, and Amortization):

This capital structure-neutral ratio reflects the company's overall valuation relative to its operating performance. It's often used in mergers and acquisitions to value a business, as it includes both equity and debt and excludes non-operating expenses.

Formula: (Market Capitalization + Debt – Cash) / EBITDA

Interpretation: A **lower** EV/EBITDA is considered more favorable, indicating better value relative to earnings.

4. PEG Ratio (Price/Earnings to Growth Ratio):

This adjusts the P/E ratio for a company's expected earnings growth, offering a more balanced view. A PEG < 1 suggests the stock may be undervalued relative to its growth, while PEG > 1 could indicate overvaluation.

Formula: P/E Ratio / Annual EPS Growth Rate

Interpretation: A **lower** PEG ratio is preferred. A value below 1 often indicates the stock may be undervalued relative to its growth potential.

5. Market Capitalization (in ₹ Crores):

Market cap represents the total market value of a company's equity. It's often used to classify companies (large-cap, mid-cap, small-cap) and indicates the scale and perceived stability of a company. Larger market caps generally imply more established companies.

Formula: Market Price per Share × Total Outstanding Shares

Interpretation: A **higher** market capitalization is generally preferred for stability, although smaller-cap firms may offer higher growth potential.

Profitability metrics:

1. Net Profit Margin

This ratio indicates the percentage of revenue that remains as profit after all expenses, taxes, and interest have been deducted. A higher net profit margin suggests stronger overall profitability and better cost control.

Formula: (Net Profit / Revenue) × 100

Interpretation: A **higher** net profit margin is preferred, as it shows better cost control and profitability.

2. Return on Equity (ROE)

ROE measures how effectively a company uses shareholders' funds to generate profits. A higher ROE indicates more efficient use of equity and is often seen as a sign of strong financial performance and management effectiveness.

Formula: (Net Income / Shareholders' Equity) × 100

Interpretation: A **higher** ROE is preferred, indicating strong returns on investor capital.

3. Return on Assets (ROA)

ROA indicates how efficiently a company uses its assets to generate profit. A higher ROA suggests better asset utilization and stronger operational efficiency.

Formula: (Net Income / Total Assets) × 100

Interpretation: A **higher** ROA is generally preferred, showing better asset utilization.

4. Gross Margin

Gross margin represents the percentage of revenue remaining after covering direct production costs. It reflects a company's pricing power and efficiency in managing production costs. Higher margins are generally favorable.

Formula: [(Revenue – Cost of Goods Sold) / Revenue] × 100

Interpretation: A **higher** gross margin is preferred, indicating stronger pricing power and production efficiency.

Efficiency metrics:

1. Asset Turnover Ratio

This ratio measures how efficiently a company uses its assets to generate revenue. A higher asset turnover indicates that the company is effectively utilizing its asset base to produce sales. It is particularly useful when comparing companies within asset-intensive industries.

Formula: Revenue / Total Assets

Interpretation: A **higher** asset turnover ratio is preferred, as it suggests more effective use of assets.

2. Inventory Turnover Ratio

This ratio reflects how quickly a company sells and replaces its inventory during a given period. A higher inventory turnover suggests efficient inventory management and strong sales performance, whereas a lower turnover may point to overstocking or weak demand.

Formula: Cost of Goods Sold (COGS) / Average Inventory

Interpretation: A **higher** inventory turnover ratio is preferred, reflecting efficient inventory management and strong sales.

Financial Health metrics:

1. Current Ratio

The current ratio measures a company's short-term liquidity—its ability to cover short-term obligations with short-term assets. A ratio above 1 indicates that the firm has more current assets than liabilities, suggesting good liquidity. However, extremely high values may indicate inefficient use of resources.

Formula: Current Assets / Current Liabilities

Interpretation: A moderately **higher** current ratio is preferred (typically between 1.5 and 2.5), indicating good liquidity without excessive idle assets.

2. Debt-to-Equity Ratio

This ratio assesses a company's financial leverage and capital structure. A higher value implies greater reliance on debt financing, which can increase financial risk. A lower ratio indicates a more conservative capital structure with less exposure to debt.

Formula: Total Debt / Shareholders' Equity

Interpretation: A **lower** debt-to-equity ratio is generally preferred, as it reflects lower financial risk and greater financial stability.

3. Interest Coverage Ratio

This ratio measures a company's ability to meet its interest obligations from operating profits. A higher interest coverage ratio indicates stronger financial health and a lower risk of default. A ratio below 1 suggests the company may not be generating enough earnings to cover its interest payments.

Formula: EBIT (Earnings Before Interest and Taxes) / Interest Expense

Interpretation: A **higher** interest coverage ratio is preferred, indicating greater capacity to meet interest obligations and lower default risk.

Return metrics:

1. Operating Margin (Operational Return)

Operating margin represents the percentage of revenue a company retains as operating profit before interest and taxes. While traditionally a profitability metric, in portfolio evaluation it can be viewed as an operational return, showing how much core profit is generated from each unit of revenue, which impacts earnings quality and stock performance.

Formula: (Operating Income / Revenue) × 100

Interpretation: A **higher** operating margin is preferred, as it reflects strong core business efficiency.

2. Dividend Yield (Shareholder Return)

Dividend yield measures the cash return an investor receives from dividends, relative to the stock's market price. It is a direct return to shareholders, especially important for income-focused investors seeking stable or growing dividend streams.

Formula: (Annual Dividend per Share / Market Price per Share) × 100

Interpretation: A **higher** dividend yield is preferred by income-seeking investors, provided it is supported by sustainable earnings.

Step 4: Metric Weight Assignment

Each metric and its sub-components were assigned weights based on their relative importance in the context of the industry. These weights were grounded in financial theory and supported by sector-specific characteristics.

Step 5:Normalization of Positively Contributing Metrics

To ensure comparability across metrics with different units and scales, we applied normalization techniques. For positively contributing indicators (e.g., Return on Equity, Net Profit Margin, Asset Turnover), the following formula was used:

$$\text{Normalized Value} = \frac{\text{Value} - \text{Min}}{\text{Max} - \text{Min}}$$

Step 6: Normalization of Negatively Contributing Metrics

For negatively contributing indicators (e.g., P/E Ratio, Debt-to-Equity Ratio, Inventory Days), we used an inverse normalization approach:

$$\text{Normalized Value} = \frac{\text{Max} - \text{Value}}{\text{Max} - \text{Min}}$$

Step 7: Weighted Score Calculation

Each normalized value was then multiplied by its respective metric weight to derive a weighted score. This ensured a balanced evaluation based on both performance and metric importance.

Step 8: Final Company Selection

The company with the highest cumulative weighted score in each industry was selected for portfolio inclusion. This indicated superior overall performance across valuation, operational efficiency, and financial strength.

Step 1: Industry Identification

Industry 1: Tele - Communications

The telecom industry plays a critical role in enabling digital infrastructure and connectivity in India, making it foundational to both economic activity and national development. With the rapid expansion of 5G, increased data consumption, and digital payments, telecom companies are set to benefit from long-term demand. Additionally, the sector features strong industry leaders with wide subscriber bases, high entry barriers, and consistent cash flows. These attributes make it attractive for stable, long-term portfolio inclusion.

We tailored the ratio weights to reflect the specific characteristics and priorities of the **telecommunications** industry, as outlined below:

Ratio Name	Weight	Reason
P/E Ratio	8.00%	Telecoms are mature and earnings-stable; P/E captures valuation based on profitability and investor sentiment.
P/B Ratio	5.00%	Book value is less indicative in telecoms due to high asset depreciation and long-lived infrastructure.
EV/EBITDA	10.00%	The most accurate metric for telecoms, accounting for debt and depreciation in this high-capex industry.
PEG Ratio	5.00%	Telecoms have stable, moderate growth; this ratio adds context to valuation by factoring in expected growth.
Market Cap (in Cr)	2.00%	While scale offers network advantages, telecom markets are often regulated, so size doesn't always mean dominance.
Net Profit Margin	7.00%	Margins matter due to high operating leverage; small margin shifts can significantly impact profits.
ROE	10.00%	Crucial for capital-heavy firms, shows how well telecoms convert invested capital into profit.
ROA	5.00%	Telecoms have large fixed asset bases; ROA shows how productively these are deployed, but leverage skews it.
Gross Margin	3.00%	Fixed infrastructure costs dominate telecoms, so gross margin varies little and has limited insight.
Asset Turnover	7.00%	Telecoms must monetize infrastructure efficiently; asset turnover captures how well they do it.
Inventory Turnover	8.00%	Many telecoms sell devices or maintain inventory; turnover reflects working capital and sales flow health.
Current Ratio	7.00%	Telecoms face high liabilities; current ratio reveals ability to meet obligations.
Debt-to-Equity	10.00%	Telecoms are often heavily leveraged due to infrastructure investment, managing debt load is critical.
Interest Coverage	3.00%	Important in debt-heavy sectors, but often less volatile in large telecoms with steady cash flows.
Operating Margin	5.00%	Reflects efficiency in core services, but overlaps with EBITDA, so slightly lower weight.
Dividend Yield	5.00%	Many telecoms are mature and offer dividends, but capital reinvestment can be more important for growth.

After finalizing the weights, we calculated the normalized values and derived the final composite scores as shown below:

	Bharti Airtel Ltd	Indus Towers Ltd	Bharti Hexacom Ltd	Tata Communications Ltd	Railtel Corporation of India Ltd
P/E Ratio	43.9	11.2	73.1	36.1	43.6
Value Weighted	0.47	1	0	0.6	0.48
P/B Ratio	10.2	3.42	16.5	15.8	6.87
Value Weighted	0.48	1	0	0.05	0.74
EV/EBITDA	14.4	6.15	24	12.7	21.8
Value Weighted	0.54	1	0	0.63	0.12
PEG Ratio	1.43	0.42	2.07	1.05	3.4
Value Weighted	0.66	1	0.45	0.79	0
Market Cap (in Cr)	1216200	111085	98005	47846	13746
Value Weighted	1	0.08	0.07	0.03	0
Net Profit Margin	0.26	0.23	0.21	0.05	0.08
Value Weighted	1	0.85	0.73	0	0.15
ROE	0.28	0.33	0.25	0.55	0.17
Value Weighted	0.3	0.44	0.23	1	0
ROA	0.06	0.17	0.07	0.05	0.07
Value	0.11	1	0.16	0	0.14

Weighted					
Gross Margin	1	0.63	1	1	1
Value Weighted	1	0	1	1	1
Asset Turnover	0.36	0.51	0.45	0.91	0.76
Value Weighted	0	0.27	0.16	1	0.73
Inventory Turnover	NA	3013	NA	3.03	NA
Value Weighted	0	0	0	0	0
Current Ratio	0.37	1.32	0.28	0.51	1.29
Value Weighted	0.09	1	0	0.22	0.97
Debt-to-Equity	1.88	0.65	1.24	4.09	0.02
Value Weighted	0.46	0.15	0.3	1	0
Interest Coverage	2.43	9.05	3.32	2.88	137
Value Weighted	0	0.05	0.01	0	1
Operating Margin	0.54	0.69	0.49	0.2	0.15
Value Weighted	0.71	1	0.63	0.08	0
Dividend Yield	0.0079	0	0.0051	0.0149	0.0067
Value Weighted	0.53	0	0.34	1	0.45
	0.418617112	**0.590529603**	0.226009585	0.523552416	0.306070556

Interpretation of the Results:

1. The weighted values for Inventory Turnover were intentionally set to zero for all companies, as the metric was not available (NA) for three of them. Including it in the weighted analysis would have introduced bias and led to an unfair comparison.
2. Based on the weighted scoring across all selected financial metrics, **Indus Towers Ltd** emerged as the top-performing stock within the **telecommunications** industry

Clarifications:

1. We didn’t include Jio in our telecom stock selection because it isn’t available as a standalone publicly traded stock, it remains part of Reliance Industries (RIL). Including it would effectively be equivalent to taking a position in RIL, which doesn't align with our goal of evaluating discrete telecom exposures.

Industry 2: Defence

The defence sector in India is gaining prominence due to increasing government focus on indigenization, rising defence budgets, and strategic geopolitical priorities. With large, long-term contracts and government backing, this sector presents a unique blend of stability and long-term growth. Prominent companies benefit from monopolistic or semi-monopolistic market positions and offer significant upside as the sector expands through Make in India initiatives.

We tailored the ratio weights to reflect the specific characteristics and priorities of the **defence** industry, as outlined below:

Ratio Name	Weight	Reason
P/E Ratio	5.00%	Earnings can fluctuate with contracts and defense budgets, so less predictive on its own.
P/B Ratio	7.00%	Important for asset-heavy defence manufacturers where tangible assets like plants and IP matter.
EV/EBITDA	12.00%	Essential for capital-intensive firms; neutralizes accounting and capex differences.
PEG Ratio	3.00%	Many defence companies grow steadily but slowly, growth potential is less central than cash flow.
Market Cap	3.00%	Larger firms win bigger contracts and offer stability, but scale isn't always correlated with performance.
Net Profit Margin	8.00%	Important for contract profitability, especially in cost-plus vs. fixed-price models.
ROE	12.00%	Vital in defence where firms must efficiently use capital while facing long sales cycles.
ROA	6.00%	Useful in assessing how well plants, equipment, and R&D investments are leveraged.
Gross Margin	5.00%	Indicates pricing power and manufacturing cost control, especially in complex production.
Asset Turnover	6.00%	Key for companies with high R&D or inventory levels; shows contract-to-revenue speed.
Inventory Turnover	5.00%	Defence has long build cycles and specialized parts, so turnover offers limited insight but still relevant.
Current Ratio	10.00%	Crucial in managing contract milestones, delayed payments, and high working capital needs.
Debt-to-Equity	12.00%	High weight due to the need for financial stability in a sector with unpredictable payment cycles.
Interest Coverage	5.00%	Important for balance sheet health, though most defense firms are cash-positive.
Operating Margin	6.00%	Contract execution efficiency is critical; reflects profitability beyond COGS.
Dividend Yield	4.00%	Defence stocks often attract long-term investors; dividends add income but are secondary to stability.

After finalizing the weights, we calculated the normalized values and derived the final composite scores as shown below:

	HindustanAeronautics Ltd	Mazagon Dock Shipbuilders Ltd	Bharat Dynamics Ltd	Cochin Shipyard Ltd	Garden Reach Shipbuilders & Engineers Ltd
P/E Ratio	39.2	53	126	63.8	65.9
Value Weighted	1	0.84	0	0.72	0.69
P/B Ratio	9.36	16.1	17.3	9.59	16.7
Value Weighted	1	0.15	0	0.97	0.08
EV/EBITDA	23.7	35.2	79.1	40.9	41
Value Weighted	1	0.79	0	0.69	0.69
PEG Ratio	1.6	1.38	229	11	2.54
Value Weighted	0.999033477	1	0	0.957736579	0.994903787
Market Cap	327559	128033	69284	53779	34739
Value Weighted	1	0.32	0.12	0.07	0
Net Profit Margin	29.00%	10.20%	15.40%	17.20%	14.90%
Value Weighted	1	0	0.28	0.37	0.25
ROE	26.10%	34.00%	14.40%	15.80%	28.10%
Value Weighted	0.6	1	0	0.07	0.7
ROA	8.93%	8.30%	4.96%	6.76%	5.12%
Value Weighted	1	0.84	0	0.45	0.04
Gross Margin	63.70%	65.70%	40.80%	60.60%	32.60%
Value Weighted	0.94	1	0.25	0.85	0
Asset Turnover	0.33	0.39	0.3	0.36	0.49
Value Weighted	0.16	0.47	0	0.32	1
Inventory Turnover	0.65	0.89	0.71	1.46	0.94
Value Weighted	0	0.3	0.07	1	0.36
Current Ratio	2.04	1.25	2.38	1.34	1.17

Value Weighted	0.72	0.07	1	0.14	0
Debt-to-Equity	0	0	0	0.09	0
Value Weighted	0	0	0	1	0
Interest Coverage	1,252	697	227	32.1	69.2
Value Weighted	1	0.55	0.16	0	0.03
Operating Margin	31.00%	18.00%	14.10%	19.30%	8.30%
Value Weighted	1	0.43	0.26	0.48	0
Dividend Yield	0.71%	0.46%	0.28%	0.48%	0.31%
Value Weighted	1	0.42	0	0.47	0.07
	0.78995702	0.527239661	0.165076031	0.575763513	0.340791913

Interpretation of the Results:

1. Based on the weighted scoring across all selected financial metrics, **Hindustan Aeronautics Ltd** emerged as the top-performing stock within the **defence** industry.

Industry 3: IT

India's IT sector is a global leader in software services, digital transformation, and outsourcing. It contributes significantly to GDP and export earnings. The sector has shown resilience during economic downturns and continues to grow through cloud services, AI, and enterprise digitalization. Market leaders in this sector deliver high return on equity, strong margins, and dividend payouts, making it an essential inclusion for both growth and stability.

We tailored the ratio weights to reflect the specific characteristics and priorities of the **IT** industry, as outlined below:

Ratio Name	Weight	Reason
P/E Ratio	14.00%	A lower P/E indicates potential undervaluation. In the margin-rich, stable IT sector, a high weight rewards firms delivering strong earnings at reasonable prices.
P/B Ratio	6.30%	IT firms are asset-light; book value may understate true worth, but it still helps identify overpriced stocks.
EV/EBITDA	7.20%	Helps normalize for varying capital structures across IT services and product companies.
PEG Ratio	4.50%	Many IT firms have moderate to high growth; PEG provides a more complete valuation picture than P/E alone.
Market Cap (in Cr)	5.00%	Larger IT firms often enjoy better client retention and pricing leverage. Scale also signals operational maturity.
Net Profit Margin (Latest Quarter)	9.00%	High profit margins are typical in IT due to low variable costs; strong margins show efficiency and pricing power.
ROE	4.20%	Useful, though less emphasized in asset-light models where equity bases are low.
ROA	3.30%	Lower relevance since intangible-heavy IT firms don't rely on physical assets.
Gross Margin (Latest Quarter GPM)	4.50%	Important to understand cost efficiency, especially for product-based or SaaS firms.
Asset Turnover Ratio	1.00%	Very low relevance in IT, since companies often have minimal physical assets and low capex.
Inventory Turnover Ratio	4.50%	Relevant mainly for hardware, product-based, or integrated IT firms (e.g., OEMs). Less so for pure services.
Current Ratio	4.50%	Moderate importance; most IT companies have healthy cash balances and low short-term liabilities.
Debt-to-Equity	12.00%	IT companies are expected to maintain low debt. High leverage in this sector is a red flag.
Interest Coverage	9.00%	Strong coverage ensures debt doesn't strain operations; highly relevant when evaluating financially disciplined IT firms.
Operating Margin (Latest Quarter)	4.50%	IT companies should maintain high operating margins to show scalability of delivery.
Dividend Yield	9.00%	Many mature IT companies (e.g., Infosys, TCS) offer consistent dividends; important for income investors.

After finalizing the weights, we calculated the normalized values and derived the final composite scores as shown below:

	Persistent Systems Ltd	Tata Consultancy Services (TCS)	Infosys	HCL Technologies
P/E Ratio	69.2	25.7	25.2	26.9
Value Weighted	0	0.9886	1	0.9614
P/B Ratio	14.7	13.1	6.97	6.71
Value Weighted	0	0.2003	0.9675	1
EV/EBITDA	43.6	17.4	15.3	16.2
Value Weighted	0	0.9258	1	0.9682
PEG Ratio	2.11	3.03	2.52	2.77
Value Weighted	0.5576	0	0.3091	0.1576
Market Cap (in Cr)	94004 Cr.	1245020 Cr.	668007 Cr.	467646 Cr.
Value Weighted	0	1	0.4978	0.3246
Net Profit Margin	0.122	0.191	0.172	0.142
Value Weighted	0.08	1	0.7467	0.3467
ROE	0.241	0.524	0.288	0.252
Value Weighted	0.0691	1	0.2237	0.1053
ROA	0.168	0.321	0.187	0.17
Value Weighted	0.0497	1	0.1677	0.0621
Gross Margin	1	1	1	0.984
Value Weighted	1	1	1	0
Asset Turnover	1.48	1.68	1.15	1.14
Value Weighted	0.6296	1	0.0185	0
Inventory Turnover	NA	2	NA	12.8
Value Weighted	0	0.0573	0	0.3668

Current Ratio	2.2	1.74	1.97	2.22
Value Weighted	0.8679	0	0.434	0.9057
Debt-to-Equity	0.05	0.1	0.09	0.09
Value Weighted	1	0	0.2	0.2
Interest Coverage	27.3	83	90.7	37.1
Value Weighted	0.0593	0.8858	1	0.2047
Operating Margin	0.18	0.263	0.241	0.214
Value Weighted	0.17	1	0.78	0.51
Dividend Yield	0.0058	0.0174	0.0267	0.0313
Value Weighted	0	0.4549	0.8196	1
	0.2601728	**0.655922**	0.6814635	0.5409388

Interpretation of the Results

1. The weights for **Inventory Turnover** were included in the analysis, as only two companies had missing (NA) values. This allowed us to retain the metric without significantly compromising the fairness of the comparison.
2. Based on the weighted scoring across all selected financial metrics, **Tata Consultancy Services (TCS)** emerged as the top-performing stock within the **IT** industry.

Clarification:

1. The industry was removed from our portfolio after we identified better-performing stocks and instruments, and to boost overall returns.
2. Its returns were very low, below the risk-free rate, resulting in a negative annual Sharpe ratio due to factors like a global slowdown in tech spending, margin pressures, and weak deal pipelines.

Industry 4: FMCG

The FMCG sector is known for its defensive nature and consistent demand, even in times of economic slowdown. It includes essential goods like food, hygiene, and personal care products. This ensures stable revenues and low volatility. Top FMCG companies benefit from powerful brand equity, strong distribution networks, and steady profitability. These characteristics make the sector ideal for balancing risk within a diversified portfolio.

We tailored the ratio weights to reflect the specific characteristics and priorities of the **FMCG** industry, as outlined below:

Ratio Name	Weight	Reason
P/E Ratio	9.00%	FMCG stocks are often richly valued due to consistent profits; P/E helps assess if they're overpriced.
P/B Ratio	6.30%	Less important in FMCG where intangible assets (brands) dominate, but still helpful in comparing capital efficiency.
EV/EBITDA	7.20%	Useful for comparing FMCG firms with different depreciation methods and capital structures.
PEG Ratio	4.50%	Growth is often modest but predictable; PEG adds perspective beyond P/E.
Market Cap (in Cr)	10.00%	Larger FMCG firms have stronger distribution networks and pricing power, size adds resilience.
Net Profit Margin (Latest Quarter)	9.00%	High margins show brand strength, pricing power, and cost control in a competitive space.
ROE	7.20%	High ROE is expected in FMCG due to asset-light, brand-rich models; vital for investor appeal.
ROA	6.30%	Important in FMCG to measure how well a company turns inventory and operations into profit.
Gross Margin (Latest Quarter GPM)	4.50%	Reflects pricing power and cost of goods sold, critical for assessing raw material impact.
Asset Turnover Ratio	4.50%	High turnover is a hallmark of FMCG; reflects operational efficiency and demand cycle speed.
Inventory Turnover Ratio	4.50%	FMCG relies on fast-moving goods; this directly ties to liquidity and demand accuracy.
Current Ratio	4.50%	Important but typically healthy in FMCG due to strong cash flows and limited receivables.
Debt-to-Equity	9.00%	FMCG companies are typically low-debt; high D/E is a red flag in an otherwise stable sector.
Interest Coverage	4.50%	Less critical but supports D/E insights; confirms stability of earnings.
Operating Margin (Latest Quarter)	4.50%	Indicates how efficiently the business turns sales into profit, key in low-margin products.
Dividend Yield	4.50%	Many FMCG stocks are favorites among income investors; consistent dividends add attractiveness.

After finalizing the weights, we calculated the normalized values and derived the final composite scores as shown below:

	Hindustan Unilever (HUL)	Nestlé India Ltd	ITC Ltd	Britannia Industries	Varun Beverages (PepsiCo bottler)
P/E Ratio	52	76.6	26.4	63.6	56.1
Value Weighted	0.49	0	1	0.26	0.41
P/B Ratio	11	57.6	7.48	32	9.4
Value Weighted	0.93	0	1	0.51	0.96
EV/EBITDA	33.8	49.3	18.2	41.3	30.5
Value Weighted	0.5	0	1	0.26	0.6
PEG Ratio	6.03	8.08	5.11	6.82	1.35
Value Weighted	0.3	0	0.44	0.19	1
Market Cap (in Cr)	542027	237057	523904	139571	156109
Value Weighted	1	0.242227722	0.95496899	0	0.041092691
Net Profit Margin(Latest Quarter)	0.164	0.16	0.275	0.126	0.131
Value Weighted	0.26	0.23	1	0	0.03
ROE	0.207	0.83	0.275	0.529	0.23
Value Weighted	0	1	0.11	0.52	0.04
ROA	0.132	0.271	0.223	0.245	0.14
Value Weighted	0	1	0.66	0.81	0.06
Gross Margin(Latest Quarter GPM)	0.514	0.562	0.588	0.401	0.546
Value Weighted	0.6	0.86	1	0	0.78
Asset Turnover Ratio	0.8	1.77	0.84	2	1.04
Value Weighted	0	0.81	0.03	1	0.2
Inventory Turnover Ratio	6.55	3.26	2.15	8.77	3.86

Value Weighted	0.67	0.17	0	1	0.26
Current Ratio	1.11	0.8	3.06	1.08	1.74
Value Weighted	0.14	0	1	0.12	0.42
Debt-to-Equity	0.03	0.28	0	0.29	0.17
Value Weighted	0.9	0.03	1	0	0.41
Interest Coverage	36.7	31.6	599	22.3	10.3
Value Weighted	0.05	0.04	1	0.02	0
Operating Margin(Latest Quarter)	0.231	0.252	0.347	0.181	0.227
Value Weighted	0.3	0.43	1	0	0.28
Dividend Yield	0.0186	0.011	0.0343	0.0127	0.0022
Value Weighted	0.51	0.27	1	0.33	0
	0.30259	0.218622772	**0.524796899**	0.17127	0.234149269

Interpretation of the Results

1. Based on the weighted scoring across all selected financial metrics, **ITC Ltd** emerged as the top-performing stock within the **FMCG** industry.

Clarification:

1. We removed FMCG from our analysis period because, despite decent fundamental ratios (as shown in the Excel sheet), it delivered negative returns.
2. This underperformance may have resulted from stocks being overpriced in April 2024 or global supply chain disruptions, driven by wars and geopolitical tensions, impacting cost structures and consumer demand.

Industry 5: Banks

The banking sector is a core component of the economy, essential for financial intermediation, credit distribution, and liquidity management. It reflects the broader economic cycle and plays a critical role in supporting growth through retail and corporate lending. We chose this sector due to its strong linkage to GDP expansion, rising credit demand, and rapid digitalization of financial services. Leading banks typically exhibit healthy balance sheets, stable net interest margins, and improving asset quality. The sector's scale, regulatory support, and consistent earnings potential make it a strategic addition to a well-diversified portfolio.

Clarification:

Banking sectors require different financial ratios because their business model, operations, and risk structures differ significantly from general industries. Traditional metrics like inventory turnover, gross margin, or current ratio don't apply meaningfully to banks, as they don't produce or sell physical goods and their "inventory" is essentially money. Instead, banks rely on interest income and financial intermediation, making ratios like Net Interest Margin (NIM), Cost-to-Income, and Loan-to-Deposit more relevant for assessing efficiency and profitability. Additionally, because banks are highly leveraged and regulated, ratios such as Capital Adequacy Ratio (CAR), Liquidity Coverage Ratio (LCR), and Gross NPA are crucial for evaluating risk, solvency, and asset quality. These specialized ratios better reflect the unique credit risk, regulatory compliance, and operational dynamics of the banking sector.

1. Net Interest Margin (NIM)

LCR assesses a bank's ability to withstand short-term liquidity disruptions by comparing its stock of high-quality liquid assets to total net cash outflows over 30 days.

Formula: (High-Quality Liquid Assets / Total Net Cash Outflows over 30 days) × 100

Interpretation: A **higher** LCR is preferred, indicating better short-term liquidity resilience.

2. Cost-to-Income Ratio (%)

The cost-to-income ratio evaluates operational efficiency by comparing operating expenses to operating income.

Formula: (Annual Dividend per Share / Market Price per Share) × 100

Interpretation: A **lower** cost-to-income ratio is preferred, as it signals higher efficiency in managing costs relative to income

3. Provision for Credit Losses (PCL) (₹ Cr)

PCL represents the funds set aside to cover potential loan defaults and credit losses in a bank's portfolio.

Formula: Provisions for Credit Losses reported in the income statement (₹ Cr)

Interpretation: A **lower** PCL is desirable, indicating better credit quality and fewer expected loan defaults

4. Debt-to-Equity Ratio

The debt-to-equity ratio measures a company's financial leverage by comparing total liabilities to shareholders' equity.

Formula: Total Debt / Shareholders' Equity

Interpretation: A **lower** debt-to-equity ratio is typically preferred, reflecting a more stable and conservative capital structure.

5. CASA Ratio (Current Account Savings Account Ratio)

CASA Ratio indicates the proportion of a bank's deposits that come from current and savings accounts, typically low-cost sources of funds.

Formula: (CASA Deposits / Total Deposits) × 100

Interpretation: A **higher** CASA ratio is preferred, as it lowers the bank's cost of funds and supports better margins.

6. Loan to Asset Ratio

This ratio measures how much of a bank's total assets are deployed in the form of loans, reflecting the lending intensity of the institution.

Formula: Total Loans / Total Assets

Interpretation: A moderately **high** loan-to-asset ratio is considered optimal, indicating effective asset utilization without overexposure to credit risk.

7. Loan-to-Deposit Ratio (LDR)

The LDR shows how effectively a bank converts its deposits into loans.

Formula: Total Loans / Total Deposits

Interpretation: A moderate LDR (typically between **70–90%**) is preferred, balancing loan growth with liquidity safety.

8. Gross NPA Ratio (Non-Performing Assets)

Gross NPA Ratio measures the percentage of total loans that are in default or close to default.

Formula: (Gross Non-Performing Assets / Gross Advances) × 100

Interpretation: A **lower** gross NPA ratio is preferred, as it reflects stronger credit risk management and better asset quality.

9. Capital Adequacy Ratio (CAR)

CAR measures a bank's capital in relation to its risk-weighted assets and current liabilities, ensuring the institution has enough buffer to absorb losses.

Formula: (Tier 1 Capital + Tier 2 Capital) / Risk-Weighted Assets × 100

Interpretation: A **higher** CAR is preferred, as it indicates financial strength and regulatory compliance.

10. Liquidity Coverage Ratio (LCR)

LCR assesses a bank's ability to withstand short-term liquidity disruptions by comparing its stock of high-quality liquid assets to total net cash outflows over 30 days.

Formula: (High-Quality Liquid Assets / Total Net Cash Outflows over 30 days) × 100

Interpretation: A **higher** LCR is preferred, indicating better short-term liquidity resilience.

We tailored the ratio weights to reflect the specific characteristics and priorities of the **Banks** industry, as outlined below:

Name of the Ratio	Weight	Reason
P/E Ratio	6%	Helpful for comparing profitability across banks, but earnings can be volatile due to provisioning swings.
P/B Ratio	8%	Most critical valuation metric for banks, assets are financial, and book value closely reflects economic reality.
PEG Ratio	5%	Adds context to P/E, though banks tend to grow steadily, not exponentially.
Market Cap	5%	Size gives stability, brand trust, and deposit base, but fundamentals remain more important.
ROE	10%	Key indicator of how well a bank uses its capital base, high ROE signals efficient lending.
ROA	7%	Important for understanding how much profit is earned from the bank's asset base (loans, securities).
Net Interest Margin (NIM)	8%	Reflects spread between lending and borrowing rates, central to banking operations.
Cost-to-Income (%)	4%	Shows how well a bank controls operating costs, lower is better, but some costs are growth investments.
Provision for Credit Losses (PCL) (Cr)	7%	Indicates how much of earnings are consumed by bad loans, rising PCL is a red flag.
Debt-to-Equity	5%	Less meaningful in banks, but still shows aggressive capital structuring.
CASA Ratio	5%	Higher CASA = lower cost of funds, giving banks pricing power and higher NIM.
Loan to Asset Ratio	5%	Indicates how much of the bank’s assets are loans (earning assets), helps assess risk and returns.
Loan-to-Deposit Ratios	5%	Higher LDR is better because it reflects more efficient use of deposits to generate income through lending, as long as it stays within safe regulatory limits.
Gross NPA Ratio	8%	Directly reflects the health of the loan book, higher NPA = higher risk of write-offs and weaker financials.
Capital Adequacy Ratio (CAR)	5%	Regulatory requirement; higher CAR = more capacity to absorb shocks and grow lending.
Liquidity Coverage Ratio (LCR)	5%	Ensures the bank can survive a short-term funding crunch, especially relevant post-2008.
Dividend Yield	4%	Useful for income, seeking investors, but less important for evaluating bank health/growth.

After finalizing the weights, we calculated the normalized values and derived the final composite scores as shown below:

	HDFC Bank	ICICI Bank	State Bank of India (SBI)	Axis Bank	Bank of Baroda
P/E Ratio	21.8	20.4	26.4	13.5	6.04
Value Weighted	0.23	0.29	0	0.63	1
P/B Ratio	2.98	3.34	1.48	2.04	0.85
Value Weighted	0.14	0	0.75	0.52	1
PEG Ratio	1.04	0.51	0.26	0.19	0.07
Value Weighted	0	0.55	0.8	0.88	1
Market Cap	1545008	1043359	718789	379913	124800
Value Weighted	1	0.65	0.42	0.18	0
ROE	14.5	18	17.2	16.4	15.5
Value Weighted	0	1	0.77	0.54	0.29
ROA	0.0174	0.0218	0.0115	0.0178	0.01
Value Weighted	0.63	1	0.13	0.66	0
Net Interest Margin (NIM)	3.13	3.83	2.5	3.37	2.56
Value Weighted	0.47	1	0	0.65	0.05
Cost-to-Income (%)	0.2916	0.3256	0.5163	0.3598	0.4794
Value Weighted	1	0.85	0	0.7	0.16
Provision for Credit Losses (PCL) (Cr)	14170	4910	19460	8170	10280
Value Weighted	0.36	1	0	0.78	0.63
Debt-to-Equity	6.46	5.97	12.4	7.48	11.1
Value Weighted	0.92	1	0	0.77	0.2
CASA Ratio	0.3818	0.4217	0.3871	0.4298	0.3781
Value Weighted	0.07	0.84	0.17	1	0
Loan to Asset Ratio	0.144	0.083	0.084	0.133	0.073
Value Weighted	1	0.14	0.15	0.85	0

Loan-to-Deposit Ratios	0.965	0.7733	0.9134	0.887	0.82
Value Weighted	1	0	0.73	0.59	0.24
Gross NPA Ratio	0.0133	0.0167	0.0182	0.0128	0.0226
Value Weighted	0.95	0.6	0.45	1	0
Capital Adequacy Ratio (CAR)	0.196	0.1655	0.1719	0.133	0.1596
Value Weighted	1	0.52	0.62	0	0.42
Liquidity Coverage Ratio (LCR)	1.1904	1.23	1.3226	1.2	1.38
Value Weighted	0	0.21	0.7	0.05	1
Dividend Yield	0.0109	0.0068	0.0197	0.0008	0.0346
Value Weighted	0.3	0.18	0.56	0	1
	0.5094	**0.6221**	0.384	0.6102	0.4065

Interpretation of the Results

1. Based on the weighted scoring across all selected financial metrics, **ICICI Bank** emerged as the top-performing stock within the Banking industry.

XAU/INR – Gold Price per Gram (INR):

Including **XAU/INR (Gold Price per Gram in INR)** in a portfolio serves as a strategic tool for diversification, risk management, and inflation protection. Gold typically exhibits a low or even negative correlation with equities and other financial assets, making it an effective diversifier that can reduce overall portfolio volatility. In times of economic uncertainty, market downturns, or geopolitical tension, gold often acts as a safe-haven asset, appreciating in value while equity markets decline. Specifically, tracking gold in INR (XAU/INR) adds another layer of protection for Indian investors by accounting for currency risk, when the Indian Rupee depreciates against the US Dollar, XAU/INR tends to rise, even if global gold prices remain flat. Additionally, gold serves as a proven hedge against inflation, preserving purchasing power when fiat currency values erode. While gold does not generate income like stocks or bonds, its stability and liquidity, especially when accessed through ETFs or Sovereign Gold Bonds, make it an efficient and flexible investment option. However, it should be used as a complementary asset rather than a primary growth engine, with a typical allocation ranging between 5% and 15% depending on the investor's risk profile and market outlook. In short, gold adds defensive strength and real asset exposure to a portfolio, helping to balance short-term volatility and long-term wealth preservation.

Final Clarifications:

1. In constructing our portfolio, we prioritized stocks with high Sharpe ratios and strong relative performance over the past year. Our primary objective was to maximize risk-adjusted returns, rather than follow popular investment themes or long-term sector narratives that lacked near-term justification.
2. While sectors such as FMCG, EV, Pharma, and Infrastructure are often considered fundamentally strong and carry long-term growth potential, they were excluded from our selection based on their weak short-term return-to-risk profiles. Despite favorable fundamentals and growth stories, many of the key players in these sectors underperformed during the time frame of our analysis.

Assignment Part A: Individual Stocks Portfolio

After a careful screening and evaluation process, we selected four assets for our portfolio: Indus Towers Ltd, Hindustan Aeronautics Ltd, ICICI Bank, and XAU/INR – Gold Price per Gram (INR). These selections were guided by a strategy focused on sectoral diversification, robust financial fundamentals, and historical performance consistency. To assess the viability of this portfolio, we conducted a one-year performance analysis using daily data from **1st April 2024 to 30th April 2025**. As part of this analysis, we calculated key performance indicators including the **Portfolio Return**, **Portfolio Risk (Standard Deviation)**, and the **Sharpe Ratio**, which collectively helped us evaluate the return-to-risk efficiency of the portfolio. This approach enabled us to make informed decisions regarding asset allocation and portfolio optimization.

Below are our analysis:

Asset	Standard Deviation	Average Return (Daily)	Annual Return
XAU/INR – Gold Price per Gram (INR)	17.00%	0.00148537476	45.36%
ICICI Bank Ltd	20.83%	0.00098332978	28.10%
Indus Towers Ltd (INDUS TOWERS.NS)	37.35%	0.00097565997	27.86%
Hindustan Aeronautics Limited (HAL.NS)	43.85%	0.00104386659	30.07%

Covariance Matrix of the portfolio:

COVARIANCE MATRIX	XAU/INR – Gold Price per Gram (INR)	ICICI Bank Ltd	Indus Towers Ltd	HAL
XAU/INR – Gold Price per Gram (INR)	0.0287908	0.001454	0.0058478	0.0050896
ICICI Bank Ltd	0.001454	0.0432257	0.0312503	0.0346954
Indus Towers Ltd	0.0058478	0.0312503	0.1384843	0.0744337
HAL	0.0050896	0.0346954	0.0744337	0.1915316

COVARIANCE MATRIX	XAU/INR – Gold Price per Gram (INR)	ICICI Bank Ltd	Indus Towers Ltd	Hindustan Aeronautics Limited
XAU/INR – Gold Price per Gram (INR)	2.88%	0.15%	0.58%	0.51%
ICICI Bank Ltd	0.15%	4.32%	3.13%	3.47%
Indus Towers Ltd	0.58%	3.13%	13.85%	7.44%
Hindustan Aeronautics Limited	0.51%	3.47%	7.44%	19.15%

Correlation Matrix of the portfolio:

CORRELATION MATRIX	XAU/INR – Gold Price per Gram (INR)	ICICI Bank Ltd	Indus Towers Ltd	Hindustan Aeronautics Limited
XAU/INR – Gold Price per Gram (INR)	1.00	0.04	0.09	0.07
ICICI Bank Ltd	0.04	1.00	0.40	0.38
Indus Towers Ltd	0.09	0.40	1.00	0.46
Hindustan Aeronautics Limited	0.07	0.38	0.46	1.00

Equally weighted portfolio, 0.25 for each:

EQUALLY WEIGHTED					
COMPANY	**XAU/INR – Gold Price per Gram (INR)**	**ICICI Bank Ltd**	**Indus Towers Ltd**	**Hindustan Aeronautics Limited**	
Weights	0.25	0.25	0.25	0.25	
Annual Return	0.453591	0.281047	0.278575	0.300719	
Standard Deviation	17.00%	20.83%	37.35%	43.85%	
PORTFOLIO	**RETURN**	**VARIANCE**	**STANDARD DEVIATION**	**STANDARD DEVIATION(%)**	**SHARPE RATIO**
VALUES	32.85%	0.04422	0.21029	21.03%	1.28

Using, Solver Tool, we optimized the portfolio:

OPTIMAL WEIGHT					
COMPANY	**XAU/INR – Gold Price per Gram (INR)**	**ICICI Bank Ltd**	**Indus Towers Ltd**	**Hindustan Aeronautics Limited**	
Weights	0.742	0.2577	-0.0048	0.005	
Annual Return	0.453591	0.281047	0.278575	0.300719	
Standard Deviation	17.00%	20.83%	37.28%	43.85%	
PORTFOLIO	**RETURN**	**VARIANCE**	**STANDARD DEVIATION**	**STANDARD DEVIATION(%)**	**SHARPE RATIO**
VALUES	40.92%	0.01929	0.13889	13.89%	2.52

Assignment Part B: Stocks Back-Testing

Back testing for these four stocks by considering one-month data **(May month 2025)** using the optimized weighs:

Covariance Matrix:

COVARIANCE MATRIX	XAU/INR – Gold Price per Gram (INR)	ICICI Bank Ltd	Indus Towers Ltd	Hindustan Aeronautics Limited
Infosys Limited	0.0595520	-0.0154818	-0.0178945	-0.0090147
ICICI Bank Ltd	-0.0154818	0.0441564	0.0127592	-0.0121413
Indus Towers Ltd	-0.0178945	0.0127592	0.0374539	0.0176522
Hindustan Aeronautics Limited	-0.0090147	-0.0121413	0.0176522	0.1162766

Optimal Weight Backtesting:

OPTIMAL WEIGHT					
COMPANY	**XAU/INR – Gold Price per Gram (INR)**	**ICICI Bank Ltd**	**Indus Towers Ltd**	**Hindustan Aeronautics Limited**	
Weights	0.742	0.2577	-0.0048	0.005	
Annual Return	0.385975	0.118191	0.164651	2.384932	
Standard Deviation	25.01%	21.53%	19.83%	34.94%	
PORTFOLIO	**RETURN**	**VARIANCE**	**STANDARD DEVIATION**	**STANDARD DEVIATION(%)**	**SHARPE RATIO**
VALUES	32.80%	0.02980	0.17263	17.26%	1.90

Assignment Part C: Indices Portfolio

Following the stock portfolio construction, we proceeded to build a second portfolio comprising four indices or mutual fund-based instruments, using the same analytical process applied earlier. Initially, we considered well-known benchmark indices such as **Nifty 50**, **Nifty 100**, **Nifty 200**, and **Nifty 500**. However, upon analyzing their one-year historical data, we observed that these indices not only delivered relatively **low returns**, but also exhibited a **high degree of correlation** with one another. This limited their effectiveness in achieving meaningful diversification and risk-adjusted returns.

As a result, we broadened our search by exploring various investment research platforms and mutual fund databases. After extensive filtering and evaluation based on historical performance, risk metrics, and diversification potential, we selected the following four instruments for portfolio construction:

- **Parag Parikh Flexi Cap Fund – Direct Growth (0P0000YWL1)**
- **Nippon India Gold Savings Fund – Direct Plan – Growth (0P0000XVDS)**
- **HDFC Flexi Cap Fund – Direct Plan – Growth Option (0P0000XW77)**
- **Aditya Birla Sun Life Credit Risk Fund – Direct Plan – Growth (0P00015W8Y)**

These funds represent a diverse mix of asset classes, including **equities**, **precious metals (gold exposure)**, and **fixed-income credit instruments**, offering broader diversification compared to traditional equity indices. We then collected one-year data **(from April 1, 2024 to April 30, 2025)** for these instruments and applied the same methodology used in the stock portfolio, calculating returns, standard deviation, and Sharpe ratio, followed by portfolio optimization using the **Solver** tool in Excel to maximize the Sharpe ratio.

Below are the analysis of these 4 indices:

Covariance Matrix of the portfolio:

Covariance Matrix Of the Indices				
COVARIANCE MATRIX	**Parag Parikh Flexi Cap Direct Growth (0P0000YWL1)**	**Nippon India Gold Savings Fund - Direct Plan - Growth (0P0000XVDS)**	**Hdfc Flexi Cap Fund -direct Plan - Growth Option (0P0000XW77)**	**Aditya Birla Sun Life Credit Risk Fund Direct Plan Growth (0P00015W8Y)**
Parag Parikh Flexi Cap Direct Growth (0P0000YWL1)	0.0125309	0.0013200	0.0129186	0.0004705
Nippon India Gold Savings Fund - Direct Plan - Growth (0P0000XVDS)	0.0013200	0.0230810	0.0016262	0.0004051
Hdfc Flexi Cap Fund -direct Plan - Growth Option (0P0000XW77)	0.0129186	0.0016262	0.0170606	0.0006091
Aditya Birla Sun Life Credit Risk Fund Direct Plan Growth (0P00015W8Y)	0.0004705	0.0004051	0.0006091	0.0019819

Covariance Matrix in %:

Covariance Matrix Of the Indices(in %)				
COVARIANCE MATRIX	**Parag Parikh Flexi Cap Direct Growth (0P0000YWL1)**	**Nippon India Gold Savings Fund - Direct Plan - Growth (0P0000XVDS)**	**Hdfc Flexi Cap Fund -direct Plan - Growth Option (0P0000XW77)**	**Aditya Birla Sun Life Credit Risk Fund Direct Plan Growth (0P00015W8Y)**
Parag Parikh Flexi Cap Direct Growth (0P0000YWL1)	1.25%	0.13%	1.29%	0.05%
Nippon India Gold Savings Fund - Direct Plan - Growth (0P0000XVDS)	0.13%	2.31%	0.16%	0.04%
Hdfc Flexi Cap Fund -direct Plan - Growth Option (0P0000XW77)	1.29%	0.16%	1.71%	0.06%
Aditya Birla Sun Life Credit Risk Fund Direct Plan Growth (0P00015W8Y)	0.05%	0.04%	0.06%	0.20%

Correlation Matrix of the Indices:

Correlation Matrix of the Indices				
CORRELATION MATRIX	**Parag Parikh Flexi Cap Direct Growth (0P0000YWL1)**	**Nippon India Gold Savings Fund - Direct Plan - Growth (0P0000XVDS)**	**Hdfc Flexi Cap Fund -direct Plan - Growth Option (0P0000XW77)**	**Aditya Birla Sun Life Credit Risk Fund Direct Plan Growth (0P00015W8Y)**
Parag Parikh Flexi Cap Direct Growth (0P0000YWL1)	1.00	0.08	0.88	0.09
Nippon India Gold Savings Fund - Direct Plan - Growth (0P0000XVDS)	0.08	1.00	0.08	0.06
Hdfc Flexi Cap Fund -direct Plan - Growth Option (0P0000XW77)	0.88	0.08	1.00	0.10
Aditya Birla Sun Life Credit Risk Fund Direct Plan Growth (0P00015W8Y)	0.09	0.06	0.10	1.00

Equally weighted portfolio, 0.25 for each:

Equally Weighted Portfolio					
COMPANY	**Parag Parikh Flexi Cap Direct Growth (0P0000YWL1)**	**Nippon India Gold Savings Fund - Direct Plan - Growth (0P0000XVDS)**	**Hdfc Flexi Cap Fund -direct Plan - Growth Option (0P0000XW77)**	**Aditya Birla Sun Life Credit Risk Fund Direct Plan Growth (0P00015W8Y)**	
Weights	0.25	0.25	0.25	0.25	
Annual Return	0.151304	0.337114	0.188151	0.182269	
Standard Deviation	11.22%	15.22%	13.09%	4.46%	
PORTFOLIO	**RETURN**	**VARIANCE**	**STANDARD DEVIATION**	**STANDARD DEVIATION(%)**	**SHARPE RATIO**
VALUES	21.47%	0.00558	0.07473	7.47%	2.09

Using, Solver Tool, we optimized the portfolio:

Optimal Weighted Portfolio					
COMPANY	**Parag Parikh Flexi Cap Direct Growth (0P0000YWL1)**	**Nippon India Gold Savings Fund - Direct Plan - Growth (0P0000XVDS)**	**Hdfc Flexi Cap Fund -direct Plan - Growth Option (0P0000XW77)**	**Aditya Birla Sun Life Credit Risk Fund Direct Plan Growth (0P00015W8Y)**	
Weights	0.054502326	0.145760496	0	0.799737178	
Annual Return	0.151304	0.337114	0.188151	0.182269	
Standard Deviation	11.22%	15.22%	13.09%	4.46%	
PORTFOLIO	**RETURN**	**VARIANCE**	**STANDARD DEVIATION**	**STANDARD DEVIATION(%)**	**SHARPE RATIO**
VALUES	20.32%	0.00195	0.04418	4.42%	3.27

Part D: Indices Back-Testing

Back testing for these four stocks by considering one-month data **(May month 2025)** using the optimized weighs:

Covariance Matrix:

Covariance Matrix Of the Indices				
COVARIANCE MATRIX	**Parag Parikh Flexi Cap Direct Growth (0P0000YWL1)**	**Nippon India Gold Savings Fund - Direct Plan - Growth (0P0000XVDS)**	**Hdfc Flexi Cap Fund -direct Plan - Growth Option (0P0000XW77)**	**Aditya Birla Sun Life Credit Risk Fund Direct Plan Growth (0P00015W8Y)**
Parag Parikh Flexi Cap Direct Growth (0P0000YWL1)	0.0124830	-0.0093043	0.0129235	0.0009424
Nippon India Gold Savings Fund - Direct Plan - Growth (0P0000XVDS)	-0.0093043	0.0347882	-0.0117799	-0.0006272
Hdfc Flexi Cap Fund -direct Plan - Growth Option (0P0000XW77)	0.0129235	-0.0117799	0.0146436	0.0011534
Aditya Birla Sun Life Credit Risk Fund Direct Plan Growth (0P00015W8Y)	0.0009424	-0.0006272	0.0011534	0.0002126

Optimal Weight Backtesting:

OPTIMAL WEIGHT					
COMPANY	**Parag Parikh Flexi Cap Direct Growth (0P0000YWL1)**	**Nippon India Gold Savings Fund - Direct Plan - Growth (0P0000XVDS)**	**Hdfc Flexi Cap Fund -direct Plan - Growth Option (0P0000XW77)**	**Aditya Birla Sun Life Credit Risk Fund Direct Plan Growth (0P00015W8Y)**	
Weights	0.054502326	0.145760496	0	0.799737178	
Annual Return	0.290333	0.094818	0.222671	0.152583	
Standard Deviation	11.46%	19.14%	12.42%	1.50%	
PORTFOLIO	**RETURN**	**VARIANCE**	**STANDARD DEVIATION**	**STANDARD DEVIATION(%)**	**SHARPE RATIO**
VALUES	15.17%	0.00070	0.02646	2.65%	5.73

Part E: Final Conclusions

Observations:

1. **Holistic and Structured Industry Evaluation**

    a. Each of the five sectors (**Telecom, Defence, IT, FMCG, Banking**) was assessed using industry-specific financial ratios with custom weightings, an advanced, thoughtful approach.
    b. The inclusion of **normalization** methods (both for positively and negatively contributing metrics) shows a clear understanding of data transformation and comparability.

2. **Transparent and Objective Stock Selection**

    a. Final stock picks, **Indus Towers, Hindustan Aeronautics, ICICI Bank, ITC, and TCS**, were based on **quantified composite scores** rather than subjective choice, demonstrating commitment to unbiased analysis.
    b. Financial data was mined, and anomalies like missing inventory turnover data were **clearly acknowledged and methodologically addressed.**

3. **Diversified Portfolio with Non-Equity Asset Inclusion**

    a. Gold (XAU/INR) was included as a **non-correlated hedge**, a strategic move that added real-asset exposure, enhanced portfolio defensiveness, and improved Sharpe Ratio.
    b. The equity-gold mix helped the portfolio benefit from **lower correlation values**, especially during backtest volatility.

4. **Empirical Backtesting Enhances Practical Insight**

    a. The backtest over May 2025 was especially useful in testing **real-world robustness**. Despite a drop in performance (Sharpe Ratio fell from 2.52 to 1.90), the portfolio remained risk-efficient, showing the natural variance between optimized theoretical portfolios and actual returns.
    b. This reflects a key lesson, optimization improves performance but cannot fully eliminate exposure to market unpredictability.

5. **Indices Portfolio Outperformed in Sharpe Terms**

    a. The **index-based mutual fund portfolio**, while more conservative, demonstrated **superior risk-adjusted returns**, particularly in the optimized version (Sharpe Ratio: 5.73).
    b. This was largely due to the high allocation to the **Aditya Birla Credit Risk Fund**, which offered **low volatility and decent yield**, a prudent choice for portfolio stability.

6. **Advanced Analytical Tools Used Effectively**

    a. Excel Solver was not just used for optimization, it was implemented with proper constraint handling, and results were interpreted with economic meaning (e.g., why negative weights appeared).
    b. Correlation and covariance matrices were used not just technically but **strategically**, to inform portfolio composition and diversification quality.

Comments:

- All relevant comments have been provided in the previous sections at intended places.

Final Conclusions:

1. **Optimization Substantially Improves Portfolio Efficiency**
    a. In both stock and index portfolios, optimized weights led to a dramatic increase in the Sharpe Ratio, from 1.28 (equal-weighted) to 2.52 (optimized) for stocks, and from 2.09 to 3.27 for indices.
    b. This clearly demonstrates the value of quantitative tools and mathematical rigor in portfolio construction.

2. **Diversification Is Necessary, But Not All Forms Are Effective**
    a. Sectoral diversification in the stock portfolio reduced correlation but introduced some volatility.
    b. Asset-class diversification (via gold and credit funds) proved more effective in managing risk, particularly in **short-term periods**, as shown by backtest performance.

3. **Real-World Testing Is Crucial**
    a. Backtesting results showed that even the best-optimized portfolios may not always maintain theoretical efficiency under live market conditions.
    b. Short-term market dynamics, sector-specific volatility (e.g., HAL or Indus Towers), and macroeconomic shifts can reduce returns, underscoring the need for active rebalancing and monitoring.

4. **Mutual Fund Instruments Can Outperform Individual Stocks in Risk-Adjusted Terms**
    a. Despite lower absolute returns, the index/mutual fund portfolio showed far lower volatility, making it ideal for conservative investors or retirement-focused portfolios.
    b. This highlights the practical merit of mutual fund strategies, especially for retail investors who lack resources for individual stock analysis.

5. **A Hybrid Strategy May Offer the Best of Both Worlds**
    a. Combining high-performing stocks (for alpha generation) with mutual funds and gold (for stability and hedging) can create a resilient, adaptable investment strategy.
    b. Future portfolios might explore this hybrid approach with more granularity, incorporating tactical shifts and macroeconomic triggers.

Tools and Sources Used

1. Microsoft Excel
2. Solver Tool
3. Screener.in
4. Investing.com
5. Groww.in
6. Yahoo Finance
